

SVALBARD, Svalbard and Jan Mayen

WMO: 010080

Lat: 78.246N

Lon: 15.466E

Elev: 88

StdP: 14.65

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	-19.3	-14.5	-25.8	1.3	-19.1	-20.6	1.8	-13.1	38.4	18.1	34.1	18.2	6.3	120	0.975

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.9	53.4	46.9	50.9	45.6	48.9	44.4	47.7	52.0	46.1	49.7	44.8	48.1	11.4	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
44.7	43.8	48.8	43.2	41.3	47.3	42.2	39.8	46.6	19.0	52.2	18.2	50.0	17.6	48.4	58.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 31.6	27.6	24.2	-17.2	58.9	9.3	3.8	-23.9	61.7	-29.3	63.9	-34.5	66.0	-41.3	68.8		
(5)			-17.6	51.2	9.1	2.6	-24.2	53.1	-29.5	54.6	-34.6	56.1	-41.2	58.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	25.4	13.4	12.1	10.3	15.9	28.2	38.8	44.8	42.9	35.7	25.8	19.7	15.6		
	DBStd	15.33	14.09	13.49	12.61	10.56	6.56	3.98	3.63	3.94	5.49	8.35	10.16	11.62		
	HDD50	9005	1135	1062	1231	1023	677	335	166	223	428	751	908	1066		
	HDD65	14471	1600	1482	1696	1473	1142	785	625	685	878	1216	1358	1531		
	CDD50	9	0	0	0	0	0	0	6	3	0	0	0	0		
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	11.1	13.3	12.8	11.9	10.1	9.5	9.6	10.2	9.0	9.5	11.4	12.5	13.5		
	PrecAvg	8.0	0.7	0.7	0.8	0.4	0.3	0.4	0.7	0.9	0.9	0.7	0.7	0.8		
	PrecMax	15.0	2.1	2.2	3.7	1.3	0.9	1.3	2.0	2.7	1.9	2.2	2.3	2.3		
	PrecMin	3.3	0.2	0.0	0.2	0.0	0.1	0.0	0.0	0.2	0.2	0.1	0.1	0.1		
	PrecStd	2.2	0.4	0.6	0.7	0.3	0.2	0.3	0.5	0.6	0.4	0.5	0.5	0.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.4	39.9	38.3	39.5	41.4	50.4	59.2	57.4	50.3	45.7	40.5	41.2		
		MCWB	36.6	36.0	34.4	35.8	36.9	43.9	49.0	47.7	46.1	42.3	37.0	37.2		
	2%	DB	37.3	37.0	35.2	36.3	39.2	47.8	54.5	52.4	47.2	42.2	38.1	36.9		
		MCWB	34.1	33.9	32.0	33.3	35.4	42.4	47.3	46.8	43.4	38.9	34.7	33.8		
	5%	DB	34.4	33.1	32.0	33.8	37.6	46.0	51.9	50.1	45.1	39.4	35.8	34.1		
		MCWB	31.4	30.1	29.7	31.3	34.3	41.2	46.0	45.3	41.7	36.7	32.9	31.3		
	10%	DB	31.6	30.4	28.3	31.2	36.1	44.4	50.0	48.3	43.0	37.0	33.2	31.7		
		MCWB	29.0	27.9	26.2	29.0	33.0	40.0	45.0	44.2	40.1	34.7	30.7	29.4		
	0.4%	WB	36.9	36.4	34.8	36.3	37.8	44.7	51.2	49.5	46.6	42.3	37.0	37.7		
		MCDB	39.9	39.4	37.6	39.2	40.2	48.4	57.0	54.4	49.9	45.2	39.8	41.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	34.3	33.7	32.1	33.7	35.9	43.0	48.4	47.3	43.8	39.2	34.9	33.9		
		MCDB	37.1	36.7	34.8	36.0	38.4	46.8	52.9	51.5	46.6	41.7	37.6	36.5		
	5%	WB	31.7	30.4	29.5	31.4	34.6	41.6	46.6	45.9	42.0	36.8	33.0	31.6		
		MCDB	34.2	33.0	31.9	33.6	37.3	45.3	50.5	49.2	44.5	39.1	35.5	34.0		
	10%	WB	28.8	27.8	26.0	29.0	33.2	40.2	45.4	44.6	40.4	34.6	30.8	29.3		
		MCDB	31.3	30.2	28.0	31.1	35.9	43.7	49.0	47.3	42.8	36.7	33.0	31.6		
	5% DB	MDBR	9.8	10.2	10.1	10.0	5.9	5.1	5.9	5.8	6.1	7.1	8.7	9.5		
		MCDBR	10.5	10.8	12.0	9.7	6.2	7.4	8.8	8.5	8.2	8.2	9.5	11.2		
		MCWBR	8.9	9.3	10.7	8.4	4.9	4.8	5.3	5.1	6.4	6.8	8.0	9.8		
		MCWBR	10.4	11.2	12.0	9.4	6.1	7.0	8.0	7.2	7.9	8.0	9.4	11.6		
	5% WB	MDBR	9.0	9.9	10.7	8.4	5.1	4.9	5.2	4.7	6.4	6.8	8.3	10.3		
		MCDBR	10.5	10.8	12.0	9.7	6.2	7.4	8.8	8.5	8.2	8.2	9.5	11.2		
		MCWBR	8.9	9.3	10.7	8.4	4.9	4.8	5.3	5.1	6.4	6.8	8.0	9.8		
		MCWBR	10.4	11.2	12.0	9.4	6.1	7.0	8.0	7.2	7.9	8.0	9.4	11.6		
(40) Clear-Sky Solar Irradiance	taub	N/A	0.187	0.244	0.278	0.298	0.292	0.297	0.286	0.237	N/A	N/A	N/A	N/A		
	taud	N/A	1.899	2.061	2.192	2.214	2.323	2.424	2.425	2.329	N/A	N/A	N/A	N/A		
	Ebn at Noon	0	10	178	245	262	272	265	246	199	0	0	0	0		
	Edh at Noon	0	1	16	26	31	29	25	21	12	0	0	0	0		
(44) All-Sky Solar Radiation	RadAvg	0	8	249	923	1557	1846	1514	949	356	0	0	0	0		
	RadStd	0	1	21	109	126	115	103	72	30	0	0	0	0		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+2.74	+8.33	+7.38	N/A	N/A	N/A	-994	-988	N/A	N/A	
(47)	Regional (5 neighbors)	+2.74	+8.33	+7.38	N/A	N/A	N/A	-994	-988	N/A	N/A	

Nomenclature: See separate page